MATH.APP.220 Multivariable Calculus

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Contents (part 2)

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Limit of a single variable real valued funtion

- On the previous courses the limit is defined by:
- The function $f : \mathbb{R} \to \mathbb{R}$ has a limit L at $a \in \mathbb{R}$, if for each $\varepsilon > 0$ there exists $\delta > 0$ (which is dependents on ε) such that

$$0 < |x - a| < \delta \quad \Rightarrow \quad |f(x) - L| < \varepsilon.$$

- The last row means that if x is close to a in the sense that the distance of x from a is at most δ, then the distance of $f(x)$ from L is at most ε.
- What should be changed if we transfer the same definition to the case of several variables?

Concepts needed for the definition

- For vectors, the equivalent of the absolute value is the **norm**

$$\|\mathbf{x}\| = \|(x_1, x_2, \ldots, x_n)\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

- The **distance** of vectors $\mathbf{x}$ and $\mathbf{a}$ is

$$\|\mathbf{x} - \mathbf{a}\| = \sqrt{(x_1 - a_1)^2 + \cdots + (x_n - a_n)^2}.$$

- Define a **neighbourhood**, also known as an **open ball**

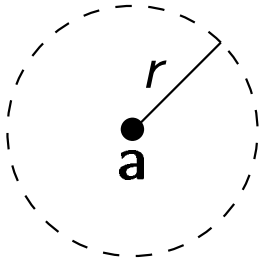
$$B(\mathbf{a}, r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{a}\| < r\}.$$

- In some circumstances you might need a **closed ball**

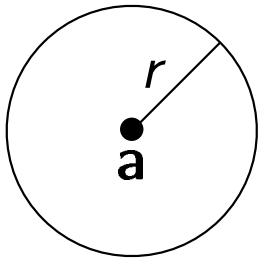
$$\overline{B}(\mathbf{a}, r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{a}\| \leq r\}.$$

A two variable neighbourhood

- In the case of two variables, the neighbourhood means a **disc** with center **a** and radius r, and whose outer boundary (a circle) is not included in the set.



open ball $B(\mathbf{a}, r)$



closed ball $\overline{B}(\mathbf{a}, r)$

- Denoting $\mathbf{x} = (x, y)$ and $\mathbf{a} = (a, b)$ the disc can be written as

$$B(\mathbf{a}, r) = \left\{ (x, y) \in \mathbb{R}^2 : \sqrt{(x-a)^2 + (y-b)^2} < r \right\}.$$

The definition of the limit for multivariable functions

- Let $D \subset \mathbb{R}^n$. The function $f : D \to \mathbb{R}$ has a **limit** L at $\mathbf{a} \in \overline{D}$, if for each $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < \|\mathbf{x} - \mathbf{a}\| < \delta \quad \Rightarrow \quad |f(\mathbf{x}) - L| < \varepsilon.$$

- This means that if the point $\mathbf{x}$ is in a neigbourhood of $\mathbf{a}$ of radius δ, then distance of $f(\mathbf{x})$ from L is at most ε.
- If the limit exists, we denote $\lim\limits_{\mathbf{x} \to \mathbf{a}} f(\mathbf{x}) = L$ or $f(\mathbf{x}) \to L$, when $\mathbf{x} \to \mathbf{a}$.
- $\lim\limits_{\mathbf{x} \to \mathbf{a}} f(\mathbf{x}) = L$ if and only if $\lim\limits_{\mathbf{x} \to \mathbf{a}} |f(\mathbf{x}) - L| = 0$

In which points does it make sense to search for limits?

- A **deleted** or **punctured** neighbourhood is one whose center is removed, $B(\mathbf{a}, r) \setminus \{\mathbf{a}\}$, where $\setminus$ is the set difference operation.
- The definition of limit contains a punctured neighbourhood, because $\|\mathbf{x} - \mathbf{a}\| > 0$. In other words the function f does not need to be defined in $\mathbf{a}$ in order to check if the limit in $\mathbf{a}$ exists.
- On the other hand, there might be some points even in the domain where it makes no sense to search for a limit.
- You have to be able to approach the point $\mathbf{a}$ along a set of points included in the domain. To put it in another way, $\mathbf{a}$ has to be in very close connection to several points of the domain.

Examples on the sensibility of trying to find a limit

- It makes sense to try to find a limit at the origin for the function
$$f(x,y,z) = \frac{xy^2z}{x^2+z^2},$$
even though the origin is not included in the domain of f. This is because all points but those on the y-axis are included in the domain.
- On the other hand, trying to find a limit at the origin for the function
$$g(x,y,z) = \ln\left(x^2+y^2+z^2-1\right)$$
makes no sense, because g is defined when $x^2+y^2+z^2>1$, i.e. outside of the closed unit ball (a ball of radius 1).

Accumulation point

- Let $D \subset \mathbb{R}^n$. The point $\mathbf{a}$ is an **accumulation point** of D, if all punctured neighourhoods of $\mathbf{a}$ (no matter how small the radius) includes points from D, in other words

$$(B(\mathbf{a}, r) \setminus \{\mathbf{a}\}) \cap D \neq \emptyset \quad \text{for all } r > 0.$$

- The accumulation points are the ones where it makes sense to try to find limits.
- Accumulations point could be describes so that they include all interior points of a set and those of the boundary points that are in immediate contact with the interior points.
- Note. Accumulation points on the boundary of D are not necessarily in D.

Properties of limits

- If $\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = L$ and $\lim_{\mathbf{x}\to\mathbf{a}} g(\mathbf{x}) = M$, then
  - Sum/difference: $\lim_{\mathbf{x}\to\mathbf{a}} (f(\mathbf{x}) \pm g(\mathbf{x})) = L \pm M$,
  - Product: $\lim_{\mathbf{x}\to\mathbf{a}} (f(\mathbf{x}) g(\mathbf{x})) = LM$,
  - Quotient: $\lim_{\mathbf{x}\to\mathbf{a}} \dfrac{f(\mathbf{x})}{g(\mathbf{x})} = \dfrac{L}{M}$, if $M \neq 0$.
- Composite function: If $\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = L$ and $\lim_{t\to L} F(t) = F(L)$, then

$$\lim_{\mathbf{x}\to\mathbf{a}} F(f(\mathbf{x})) = F\left(\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x})\right) = F(L).$$

Hints on calculating limits

- If you can evaluate the function at the limit point **a** without problems, i.e. the the function is continuous, just insert the coordinates of **a** into the corresponding variables.
- If this kind of substitution causes indeterminate forms like $\frac{0}{0}$ to appear, use one of the techniques (introduced on previous courses) to get rid of these forms, for example
  - reduction (if you take a common factor from the denominator, does the numerator have the same common factor?)
  - expansion (for example, expanding a fraction containing $\sqrt{f(x,y)} - \sqrt{g(x,y)}$ by $\sqrt{f(x,y)} + \sqrt{g(x,y)}$)

How to prove the limit does not exist

- In case of a single variable function, you can only approach the limit point from two directions. The limit exists if the one-sided limits are equal. If not, there's no limit.
- In case of a multivariable function, the point **a** can be approached from infinitely many directions.
- If you find two directions that give different limits when approaching **a**, there is no limit at **a**.
- Note! If the limits from two directions are equal, this does NOT prove there is a limit. You might have just accidentally found two directions that give the same limit value, but other directions would give other limits.

Approaching a point from different directions

- Point **a** can be approached along a curve (for example a line).
- Calculate a limit along the curve by substituting x and y with the equation of the curve so that a single variable function results. Calculate its limit like you did on previous courses.
- Example. Consider the limit of the function $f(x,y) = \dfrac{xy}{x^2+y^2}$ at the origin. Approach the origin along the line $y = x$. On this line, when $x \neq 0$, f is a constant function,

$$f(x,x) = \frac{x \cdot x}{x^2+x^2} = \frac{x^2}{2x^2} = \frac{1}{2}.$$

The limit of this function at the origin (i.e. when $x \to 0$) is $\frac{1}{2}$.

Approaching a point from different directions

- Continuing with the limit example, next approach the origin along the y-axis (i.e. $x = 0$). On this line, when $y \neq 0$,

$$f(0, y) = \frac{0 \cdot y}{0^2 + y^2} = \frac{0}{y^2} = 0.$$

  The limit of this function at the origin (i.e. when $y \to 0$) is 0.

- Because the limits approaching the origin along two curves were not the same, the function f has no limit at the origin.

- Note that even though the above function along the y-axis was $f(0, y) = \frac{0}{y^2}$, we don't get $\frac{0}{0}$ when $y \to 0$. Finding a limit means focusing on how the function behaves when the limit point is not reached yet. On the y-axis outside the origin it is allowed to simplify $\frac{0}{y^2} = 0$.

Infinite limit

- If the limit does not exist due to the value of the function growing without bound when **a** is approached from any direction, we say that f has an **infinite limit** ∞ at **a**. If $-f$ grows without bound, the limit of f at **a** is $-\infty$. For example,

$$\lim_{(x,y)\to(0,0)} \frac{1}{x^2+y^2} = \infty.$$

- Remember that the limit might not exist for other reasons. The limit

$$\lim_{(x,y)\to(0,0)} \sin\left(\frac{1}{x^2+y^2}\right),$$

does not exist, because the value of the function oscillates when approaching the origin.

Definition of continuity

- Let $D \subset \mathbb{R}^n$ and $\mathbf{a}$ is an accumulation point of D. The function $f : D \to \mathbb{R}$ is **continuous at** $\mathbf{a} \in \mathbb{R}^n$, if

$$f(\mathbf{a}) = \lim_{\mathbf{x} \to \mathbf{a}} f(\mathbf{x}).$$

- For non-accumulation points $\mathbf{a} \in D$, f is continuous at $\mathbf{a}$.
- If f is continuous in all points of D, f is **continuous**.
- Note that the equation in the definition contains three things:
  - the function f is defined at $\mathbf{a}$
  - the function f has a limit at $\mathbf{a}$
  - the value $f(\mathbf{a})$ and the limit of f at $\mathbf{a}$ are equal

Removable discontinuity

- A point of discontinuity is called **removable**, if the function f is discontinuous at $\mathbf{a}$, but $\lim\limits_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x})$ exists.
- For example, $f(x,y) = \dfrac{\sin\left(x^2+y^2\right)}{x^2+y^2}$ is not defined at the origin, and therefore discontinuous, but $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 1$. Let's "remove" the discontinuity by defining the function

$$g(x,y) = \begin{cases} \dfrac{\sin\left(x^2+y^2\right)}{x^2+y^2}, & \text{if } (x,y) \neq (0,0), \\ 1, & \text{if } (x,y) = (0,0). \end{cases}$$

Reasoning when justifying a function is continuous

- Let $D \subset \mathbb{R}^n$ and $\mathbf{a}$ is an accumulation point of D. If the functions $f, g : D \to \mathbb{R}$ are continuous at $\mathbf{a}$, then also the following functions are continuous at $\mathbf{a}$:
  - Sum/difference: $f(\mathbf{x}) \pm g(\mathbf{x})$,
  - Product: $f(\mathbf{x})\, g(\mathbf{x})$,
  - Quotient: $\dfrac{f(\mathbf{x})}{g(\mathbf{x})}$, if $g(\mathbf{a}) \neq 0$.
- If additionally the function $F : \mathbb{R} \to \mathbb{R}$ is continuous at $f(\mathbf{a})$, then the composite function $(F \circ f)(\mathbf{x})$ is continuous at $\mathbf{a}$.
- In general, you may justify the continuity of functions in a similar manner as you did on for single variable functions on a previous course.

Multivariable polynomials and rational functions

- The function $P : \mathbb{R}^n \to \mathbb{R}$ is a **polynomial** of degree k, if it is of the form

$$P(\mathbf{x}) = \sum_{k_1+\cdots+k_n \leq k} c_{k_1,\ldots,k_n} x_1^{k_1} \cdots x_n^{k_n},$$

where $c_{k_1,\ldots,k_n} \in \mathbb{R}$ and $k_i \in \{0,1,2,\ldots,k\}$ for $i = 1,2,\ldots,n$.

- An example of a polynomial of degree 5 (on which the first two terms are of degree 5)

$$P(x,y,z) = 2x^2yz^2 + 3xy^4 - 5y^2z + 2xyz - 6z^3 - 2x + 3$$

- A **rational function** is of the form $R(\mathbf{x}) = \dfrac{P(\mathbf{x})}{Q(\mathbf{x})}$, where P and Q are polynomials.

The definition of a partial derivative

- Unlike in the case of a single variable function, we can now differentiate with respect to more than one variable. That's why the term partial derivative.
- For a two variable function $f : D \to \mathbb{R}$, the **partial derivative** with respect to x at (a, b) is

$$\frac{\partial}{\partial x} f(a, b) = \lim_{h \to 0} \frac{f(a+h, b) - f(a, b)}{h},$$

  if the limit exists.
- Similarly, the partial derivative with respect to y at the same point is

$$\frac{\partial}{\partial y} f(a, b) = \lim_{k \to 0} \frac{f(a, b+k) - f(a, b)}{k}.$$

Notations and calculation of partial derivatives

- The following notations are used for partial derivatives:

$$\frac{\partial}{\partial x} f(x,y) = \frac{\partial f}{\partial x}(x,y) = D_x f(x,y) = f_x(x,y),$$
$$\frac{\partial}{\partial y} f(x,y) = \frac{\partial f}{\partial y}(x,y) = D_y f(x,y) = f_y(x,y).$$

- When you calculate partial derivatives, use the same differentiation rules as for single variable functions, but so that when you differentiate with respect to one variable, but assume that the rest of the variables are constants.
- For example, when differentiating with respect to y, assume x is constant

$$\frac{\partial}{\partial y}\left(2x^2 - 3xy + y^2\right) = 0 - 3x + 2y = -3x + 2y.$$

Geometric interpretation

- Geometrically the partial derivative $\frac{\partial f}{\partial x}(a,b)$ represents the slope of the tangent line to the intersection curve denoted by $z = f(x,b)$ of the surface $S = \{(x,y,z) : z = f(x,y)\}$ and the plane $y = b$ at the point $(a,b,f(a,b))$.
- The equation of the tangent line to the curve of intersection $z = f(x,b)$ is

$$z = f(a,b) + \frac{\partial f}{\partial x}(a,b)(x-a)$$

- Similarly, $\frac{\partial f}{\partial y}(a,b)$ is the slope of the tangent line to the intersection curve $z = f(a,y)$. The tangent line is given by

$$z = f(a,b) + \frac{\partial f}{\partial y}(a,b)(y-a)$$

Tangent plane

- Two non-parallel lines or vectors span a plane (known from a previous course)
- The equation of a plane through the point $(a, b, f(a, b))$ is

$$z = f(a, b) + A(x - a) + B(y - b),$$

  where A and B are constants that tell about the inclination (tilt) of the plane.
- Two tangent lines at $(a, b, f(a, b))$ also span a plane. It is the **tangent plane** to the surface S at $(a, b, f(a, b))$.
- Comparing with equations of the tangent lines, we notice that the equation of the plane must be

$$z = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b).$$

Critical points

- The point (a, b) is a **critical point** of the function $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^2$, if both partial derivatives vanish (=are zero) at (a, b), or at least one of the partial derivatives does not exist in (a, b).
- If (a, b) is a critical point of the type where partial derivatives vanish, then the tangent plane to the surface $S=\{(x, y, z) : z = f(x, y)\}$ at $(a, b, f(a, b))$ is horizontal and given by the equation $z = f(a, b)$.
- The critical points are important when finding extrema. We will return to then later in the course.

The definition of a partial derivative

- The **partial derivative** with respect to x_i of a function $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, at $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is

$$\begin{aligned}\frac{\partial}{\partial x_i} f(\mathbf{x}) &= \lim_{h \to 0} \frac{f(x_1, \ldots, x_{i-1}, x_i + h, x_{i+1}, \ldots, x_n) - f(x_1, \ldots, x_n)}{h} \\ &= \lim_{h \to 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h}\end{aligned}$$

  if the limit exists.

- The latter form of this limit contains the vector $\mathbf{e}_i$ that is a unit vector in the direction of the ith coordinate. For example $\mathbf{e}_4 = (0, 0, 0, 1, 0)$ in $\mathbb{R}^5$. Therefore, also in the limit on the second row, the value h is added to the ith coordinate of the vector $\mathbf{x}$.

Existence of partial derivatives

- Because the ith coordinate is increased by the value h in the definition of the partial derivative, at the points at the boundary of the domain it is possible to end up at a point where f in not defined.
- Therefore the partial derivatives exist only in the interior points of the domain, where a (small) addition of h always does not take us away from the domain.
- A point $\mathbf{x} \in D$ is an **interior point** of $D \subset \mathbb{R}^n$, if for at least one radius $r > 0$ it holds that $B(\mathbf{x}, r) \subset D$.
- The partial derivatives are generally calculated the same way as for two variable functions, i.e. assuming all other variables than the differentiation variable are constant.

Second order partial derivatives

- Consider the function $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, all of whose partial derivatives $\frac{\partial f}{\partial x_i}$ $\forall i = 1, 2, \ldots, n$ exist in a neighbourhood of $\mathbf{x}$. Then the partial derivatives of the functions $\frac{\partial f}{\partial x_i}$ are called the **second order partial derivatives** of f and denoted as

$$\frac{\partial}{\partial x_j}\left(\frac{\partial f}{\partial x_i}\right)(\mathbf{x}) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}) = (f_{x_i})_{x_j} = f_{x_i x_j} = D_j D_i f.$$

- In the notation $f_{x_i x_j}$, the differentiation order is from left to right, and in the notations $D_j D_i f$ and $\frac{\partial^2 f}{\partial x_j \partial x_i}$ from right to left.
- In short we can write $\dfrac{\partial^2 f}{\partial x_i \partial x_i} = \dfrac{\partial^2 f}{(\partial x_i)^2} = \dfrac{\partial^2 f}{\partial x_i^2}$.

Equality of mixed partials

- If we say just "all partial derivatives", we mean all first order partials. For other orders the ordinal number is mentioned.
- For the so called mixed partial derivatives of second or higher order the following equality holds.
- If $\frac{\partial^2 f}{\partial x_i^2}$, $\frac{\partial^2 f}{\partial x_j^2}$ and $\frac{\partial^2 f}{\partial x_j \partial x_i}$ exist and $\frac{\partial^2 f}{\partial x_j \partial x_i}$ is continuous in the neighbourhood of $\mathbf{x}$, then $\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x})$ exists and

$$\frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}) = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}).$$

- For example, $f_{xxy} = f_{xyx} = f_{yxx}$, if this partial derivative is continuous.